南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY
电子与电气工程系
DEPARTMENT OF ELECTRONIC AND ELECTRICAL ENGINEERING

Fundamentals of Optics

Lecturer

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Q&A time: 2-4 pm on Monday

Room 239, South Tower, College of Engineering

Rules

- **Final exam: 30%**
- **Mid-term exam: 30%**
- **Homework: 25%**
- **Attendance: 15%**

Reference books

Optics, by Eugene Hecht
光学，赵凯华、钟锡华

CHAPTER 4
GEOMETRICAL OPTICS

4.1 Introductory Remarks

A point from which a portion of a spherical wave diverges, or one toward which the wave segment converges, is known as a focus of the bundle of rays.

Object space

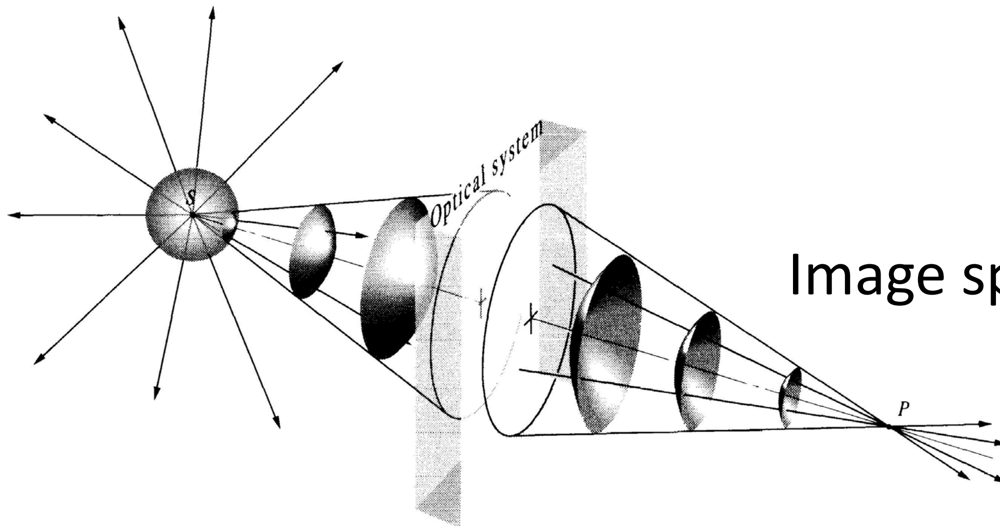


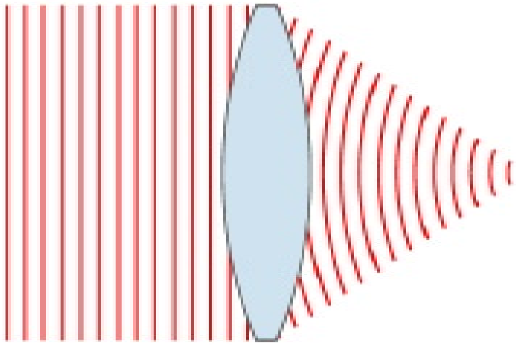
Image space

Point S and P are spoken of as conjugate points

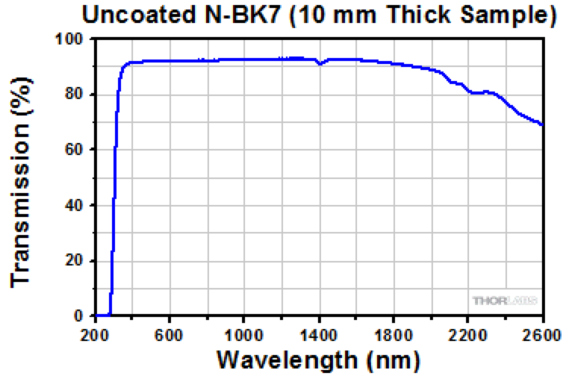
Geometrical Optics: as the wavelength of the radiant energy decreases in comparison to the physical dimensions of the optical system, the effects of diffraction become less significant. Point S and P are spoken of as conjugate points

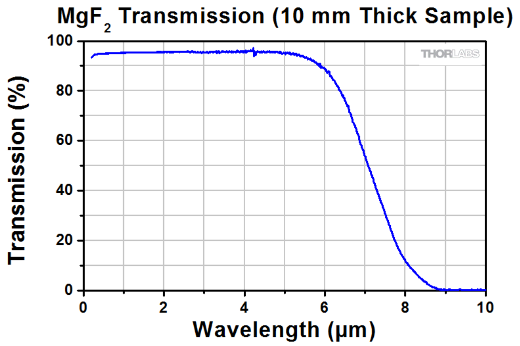
4.2 Lenses

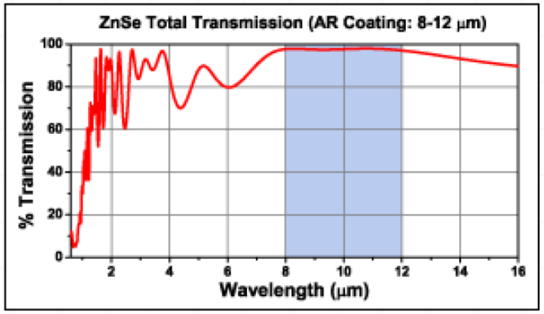
A lens is a refracting device that reconfigures a transmitted energy distribution.

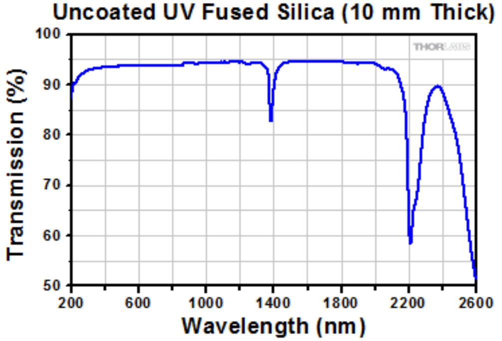


For UV, lightwave, IR, microwave









4.2.1 Aspherical Surfaces

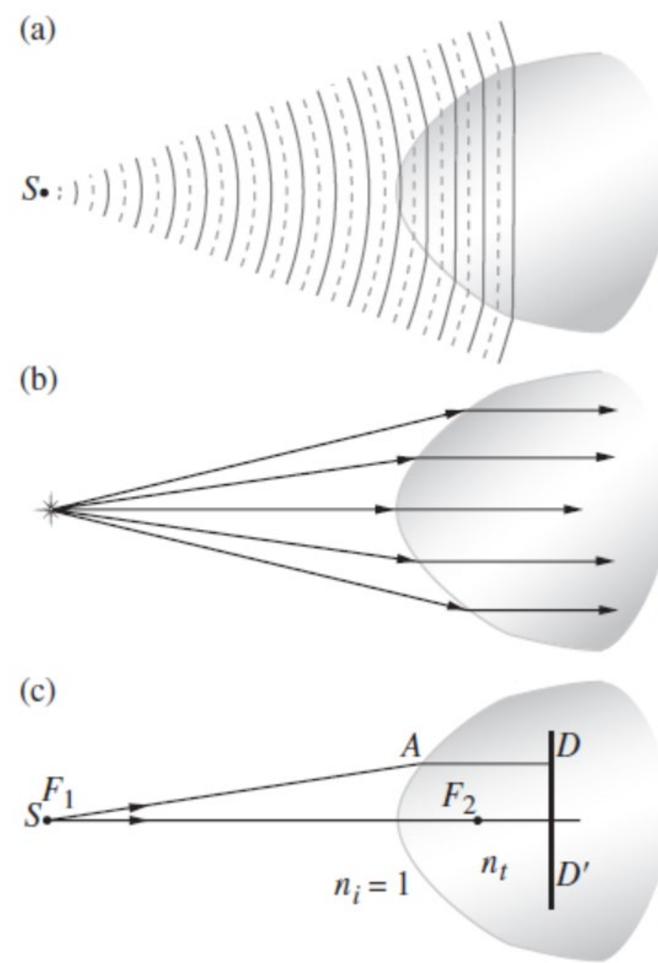
Whatever path the light takes from S to $\overline{DD'}$, it must always be the same number of wavelengths long, so that the disturbance begins and ends in-phase.

$$\overline{F_1 A}/\lambda_i + \overline{AD}/\lambda_t = C_1$$

$$n_i\left(\overline{F_1 A}\right) + n_t\left(\overline{AD}\right) = C_2$$

$$\overline{F_1 A}/v_i + \overline{AD}/v_t = C_3$$

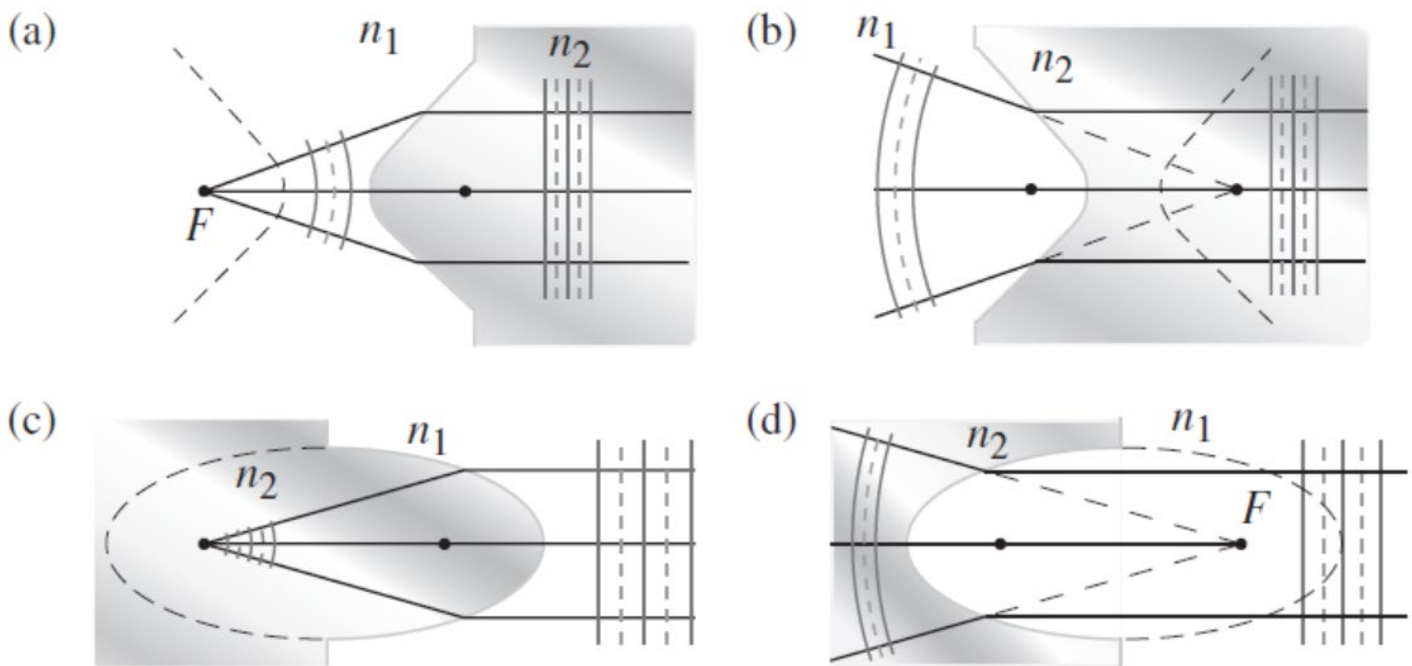
All paths from S to $\overline{DD'}$ must take the same amount of time to traverse.



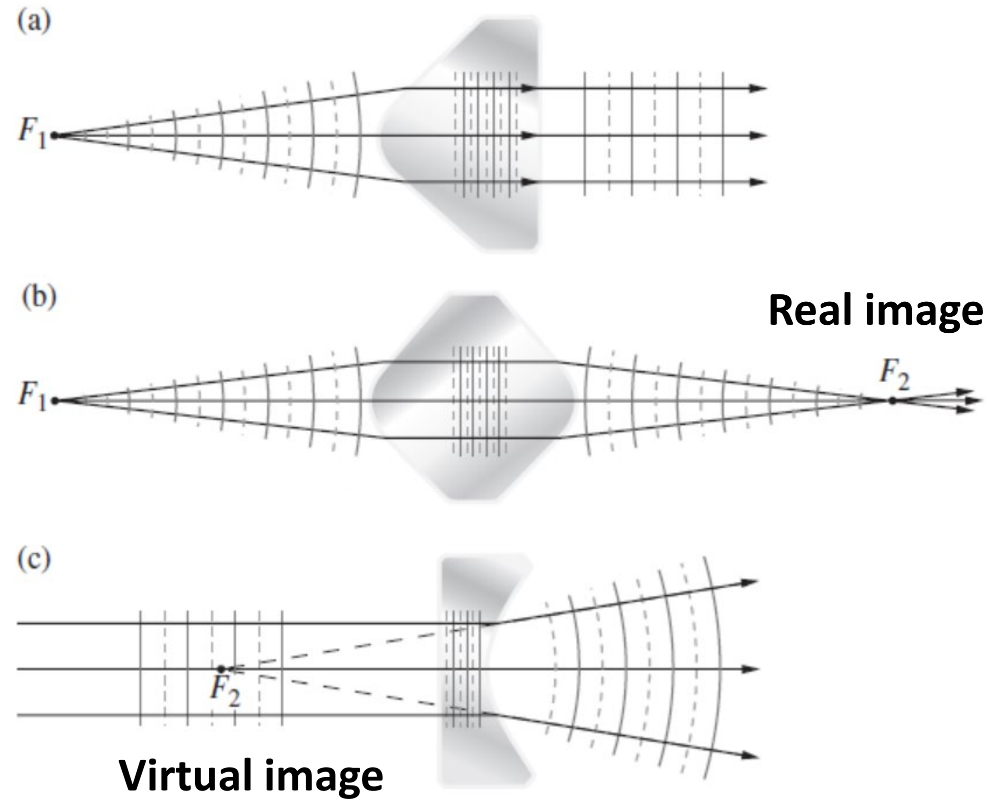
$$(\overline{F_1 A}) + \frac{n_t}{n_i}(\overline{AD}) = constant$$

Eccentricity (偏心率): $e = n_{ti} > 1$

The greater the eccentricity, the flatter the hyperbola (双曲线).

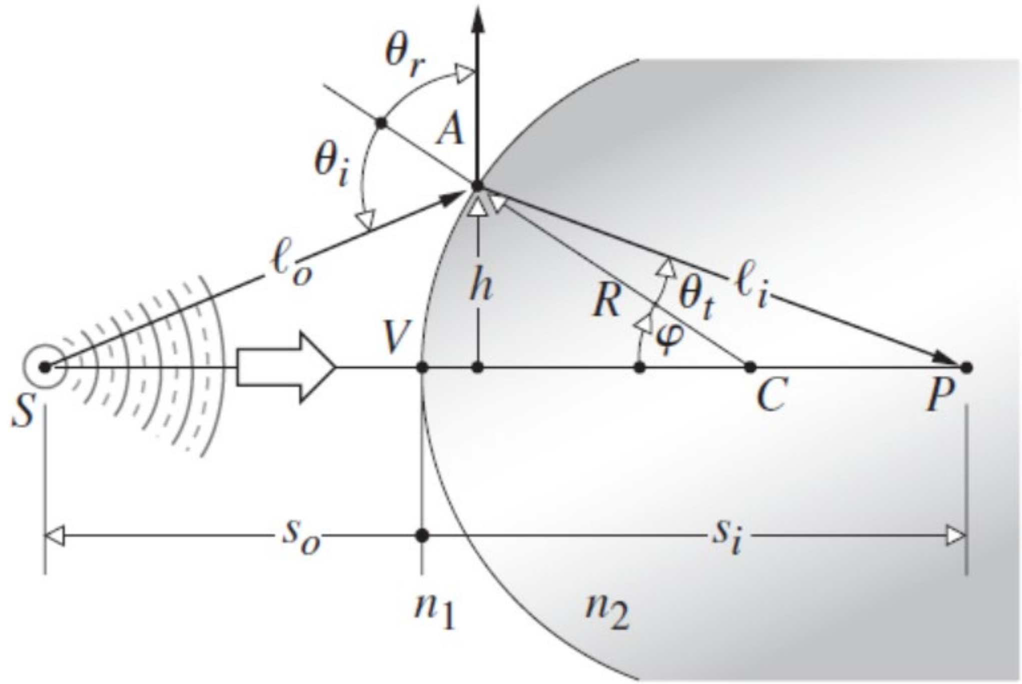


- **Convex**: the lenses are thicker at their midpoints than at their edges.
- **Converging lenses**: each lens causes the incoming beam to converge somewhat, to bend a bit more toward the central axis.
- **Concave**: the lenses are thinner in the middle than at the edges.
- **Diverging lenses**: these devices turn rays outward away from the central axis.



4.2.2 Refraction at Spherical Surfaces

Vertex: the point V
Objective distance: $s_o = \overline{SV}$
Image distance: $s_i = \overline{VP}$



Fermat's Principle:

$$\frac{d(OPL)}{d\varphi} = 0$$

where $OPL = n_1\ell_0 + n_2\ell_i$

We have

$$\frac{n_1}{\ell_0} + \frac{n_2}{\ell_i} = \frac{1}{R}\left(\frac{n_2 s_i}{\ell_i} - \frac{n_1 s_o}{\ell_0}\right)$$

First-order theory:

$$\cos\varphi = 1 - \frac{\varphi^2}{2!} + \frac{\varphi^4}{4!} - \frac{\varphi^6}{6!} + \cdots$$

$$\sin\varphi = \varphi - \frac{\varphi^3}{3!} + \frac{\varphi^5}{5!} - \frac{\varphi^7}{7!} + \cdots$$

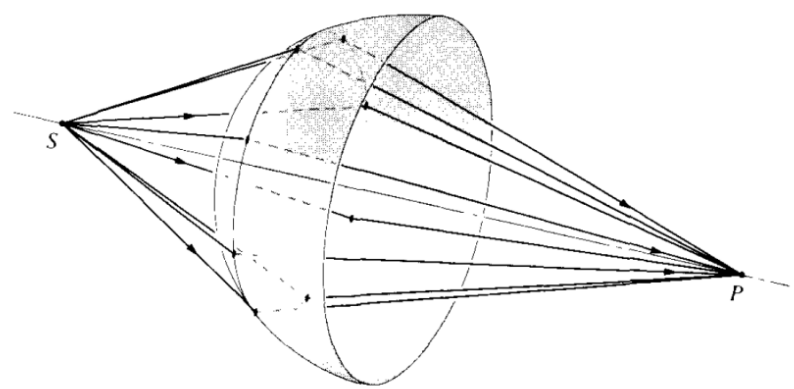
$$\frac{n_1}{\ell_0} + \frac{n_2}{\ell_i} = \frac{1}{R}\left(\frac{n_2 s_i}{\ell_i} - \frac{n_1 s_0}{\ell_0}\right)$$

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

Paraxial (傍轴) rays: rays that arrive at shallow angles with respect to the optical axis.

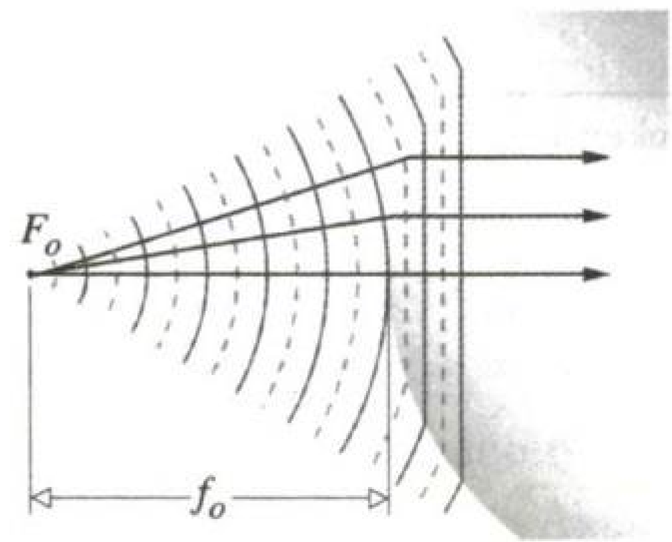
Same meaning: first-order, paraxial, or Gaussian optics

Paraxial region: the emerging wavefront segment corresponding to these paraxial rays is essentially spherical and will form a "perfect“ image at its center p located at s_i.



When $s_i = \infty$

$$\frac{n_1}{s_o} + \frac{n_2}{\infty} = \frac{n_2 - n_1}{R}$$

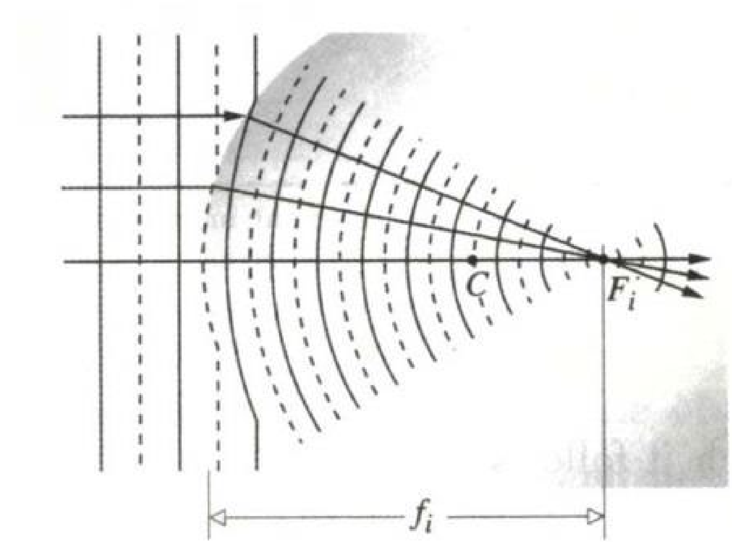


First or objective focal length: $s_o \equiv f_0$

$$f_0 = \frac{n_1}{n_2 - n_1} R$$

When $s_o = \infty$

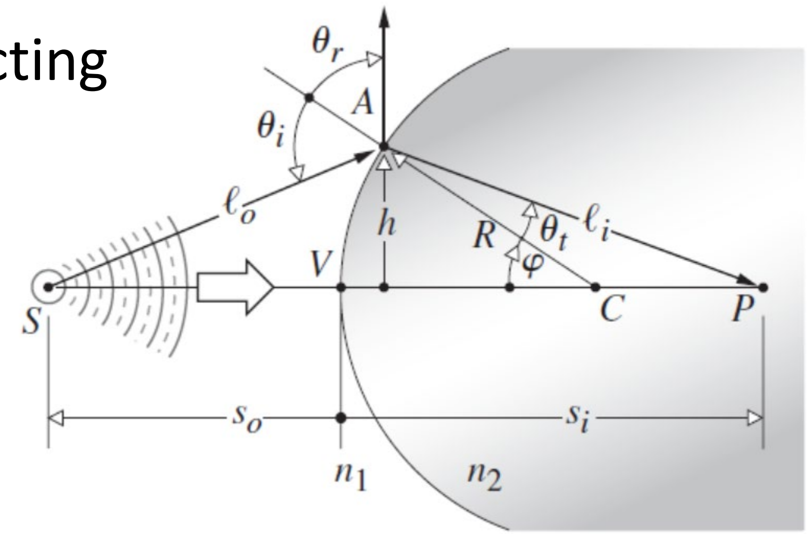
$$\frac{n_1}{\infty} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$



Second or image focal length: $s_i \equiv f_i$

$$f_i = \frac{n_2}{n_2 - n_1} R$$

Sign convention for spherical refracting surfaces and thin lenses (light entering from the left)

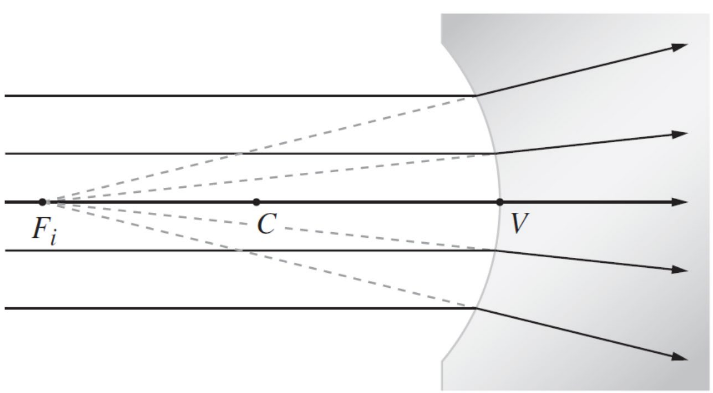


$s_o,\ f_0$: + left of V

$s_i,\ f_i$: + right of V

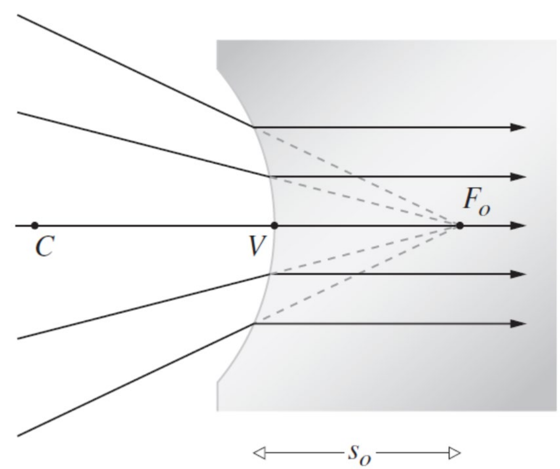
R : + if C is right of V

R, f_i, s_i: negative



An image is virtual when the rays diverge from it.

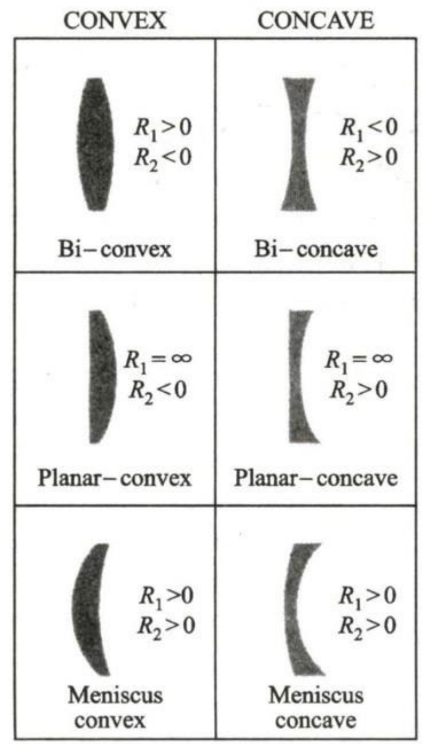
s_o, R, f_0, s_i: negative

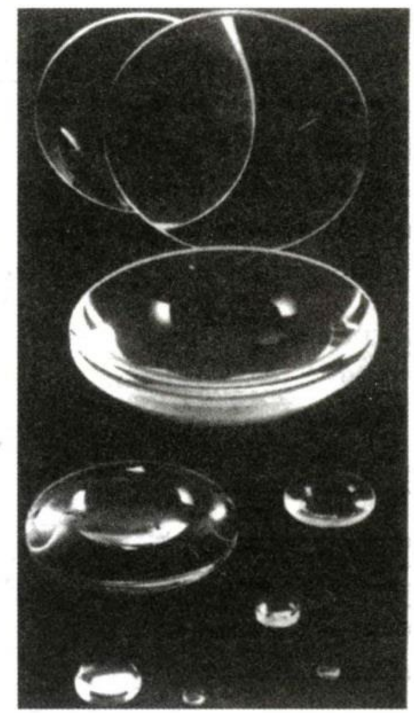


An object is virtual when the rays converge toward it.

4.2.3 Thin lenses

- Convex, converging, or positive lenses: thick at the center and so tend to decrease the radius of curvature of the wavefronts. (index of the lens is greater than that of the media)
- Concave, diverging, or negative lenses: thin at the center and tend to advance that portion of wavefront, causing it to diverge more than it did upon entry.



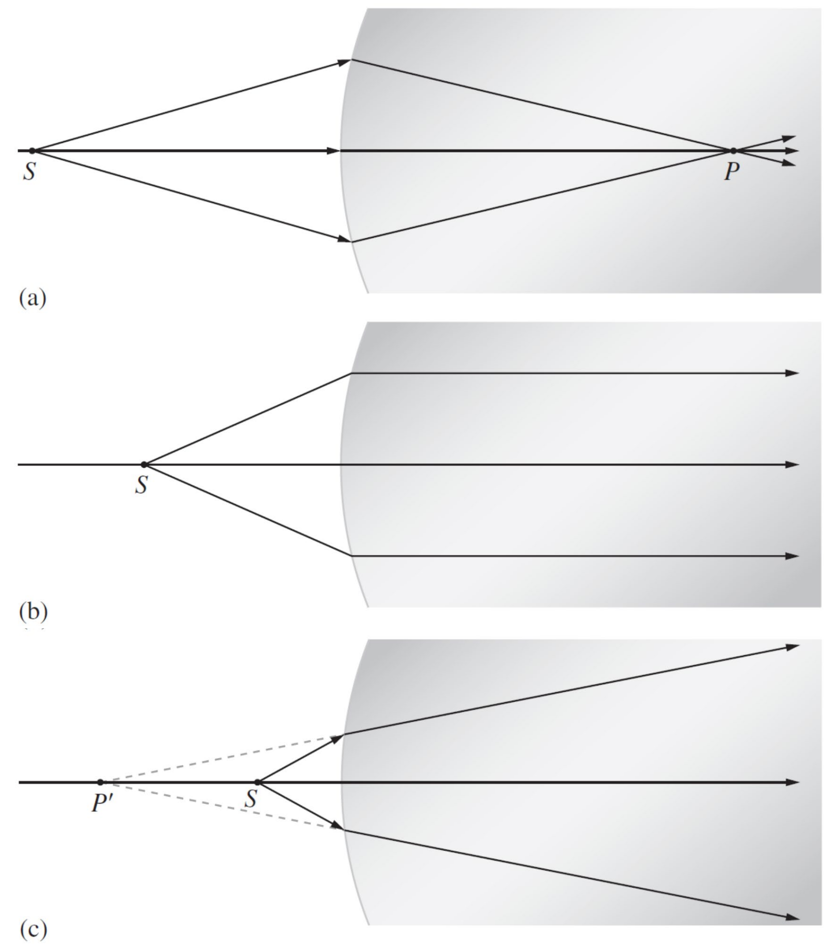


In the paraxial region, the location of the conjugate points S and P is given by

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$

As s_o decreases, s_i moves away from the vertex.
Finally, $s_o = f_0$, $s_i = \infty$.

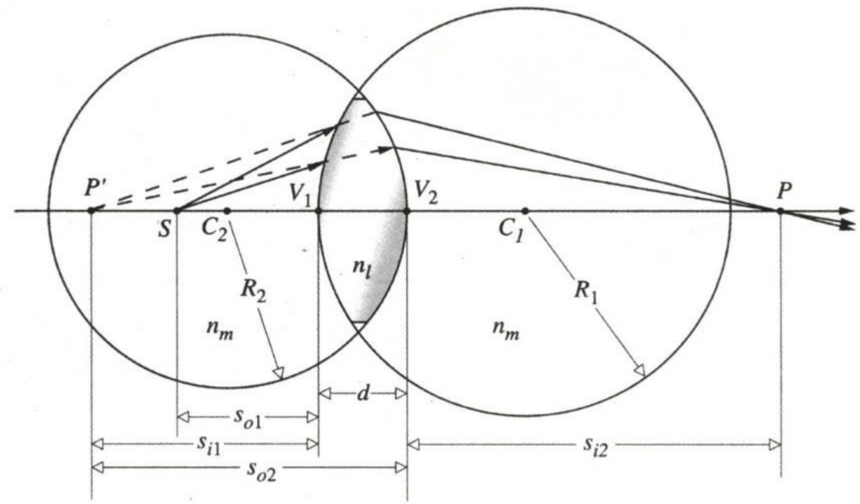
If s_o gets any smaller, s_i will have to be negative.
In other words, the image becomes virtual.



Thin-lens equations

In the paraxial region

$$\frac{n_1}{s_o} + \frac{n_2}{s_i} = \frac{n_2 - n_1}{R}$$



At the first surface, paraxial rays issuing from S at s_{o1} will meet at P', a distance, which we now call s_{i1}, from V_1, given by

$$\frac{n_m}{s_{o1}} + \frac{n_l}{s_{i1}} = \frac{n_l - n_m}{R_1}$$

At the second surface:

$$\frac{n_l}{-s_{i1} + d} + \frac{n_m}{s_{i2}} = \frac{n_m - n_l}{R_2}$$

Add above equations, we have

$$\frac{n_m}{s_{o1}} + \frac{n_m}{s_{i2}} = (n_l - n_m)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) + \frac{n_l d}{(s_{i1} - d)s_{i1}}$$

For thin lens ($d \to 0$) surrounding by air ($n_m \approx 1$)

Thin-lens equation

$$\boldsymbol{\frac{1}{s_o} + \frac{1}{s_i} = (n_l - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$